

Table 1: Milestone progress for the CAN-FD bus transceiver implementation. Milestones are form the implementation road map provided by FKRI.

#	Milestone	Status	Progress	Est. Remaining	Notes
1	CAN MAC TX (no error handling)	Done (review pending)	100%	-	TBs: <code>can_mac_tx_tb</code> , <code>can_mac_ser_tx_tb</code> , <code>can_mac_bs_tb</code> , <code>can_mac_crc_tb</code> .
2	CAN MAC RX (no error handling)	Done (review pending)	100%	-	TB: <code>can_mac_rx_tb</code> .
3	TB CAN MAC RX-TX (no error handling)	Skipped	-	-	TX and RX share no state at the MAC level. The <code>can_mac</code> wrapper merges their FCE signals with a simple OR, which is verified structurally. Each path is fully exercised by its own TB. I don't think a combined TB would cover additional logic.
4	CAN FCE + MAC error handling	Done (review pending)	100%	-	Integrated into <code>can_mac</code> wrapper. TB: <code>can_fce_tb</code> .
5	Extend TB with MAC error cases	Partial	60%	2 days	TX done (8 injection types in <code>can_mac_tx_tb</code>). RX error injection not yet implemented.
6	CAN PCS RX TX (no error handling)	Not started	0%	5 days	TX and RX PCS modules needed.
7	Extend TB with PCS timing	Not started	0%	3 days	Depends on milestone 6.
8	CAN FCE PCS error handling	Done (review pending)	100%	-	Bus-off entry/recovery tested in <code>can_fce_tb</code> .
9	Extend TB with PCS functionality	Not started	0%	3 days	Bit timing, TDC, SP/SSP verification. Depends on 6 and 7.
10	CAN LLC RX TX (no error handling)	Not started	0%	5 days	TX and RX LLC modules needed.
11	Extend TB with LLC	Not started	0%	3 days	Depends on milestone 10.
12	CAN FCE LLC error handling	Done (review pending)	100%	-	Bus-off indication and recovery tested in <code>can_fce_tb</code> .

#	Milestone	Status	Progress	Est. Remaining	Notes
13	Extend TB with LLC error cases	Not started	0%	5 days	Depends on milestones 11 and 12.
-	Documentation (report, verification plan)	Ongoing	-	Continuous	
Sum				16 days	

Table 2: Testbench execution status and affirmation counts.

Testbench	Affirmations	Coverage	Status
can_fce_tb	227	Counter updates, state transitions, bus-off recovery	Passing
can_mac_bs_tb	11,561	Reset (252), dynamic stuff (208), FSB directed (11), SBC (11,076), FSB (12)	Passing
can_mac_crc_tb	54,084	Reset (1,008), CRC check (1,002), output stable (52,074)	Passing
can_mac_ser_tx_tb	793,708	Coverage-driven: IDE, FDF, DLC 0-15	Passing
can_mac_tx_tb	268,903	PCS VC (264,445), transfer status (2,222), FCE events (2,222), reset (14). Coverage: IDE, FDF, FTYP, ESI, BRS, DLC, error injection type/position, FCE state	Passing
can_mac_rx_tb	14,704	Frame check (14,701), reset (3). Coverage: IDE, FDF, DLC	Passing